

1.a. Standard deviation is the spread of an observed variable around its mean. Standard Error is the standard deviation in the sampling distribution.

b. Randomization inference assumes that there is no effect. Under this assumption it creates n-repetitions of the experiment, where the effect is randomly shuffled and for every combination the difference in means (for no effect and effect) is calculated. If there is no treatment effect, the random shuffling should make no difference. Then you compare the results with the observed result. The p value is what percentage of the random samples produced a result at least as extreme as my observed value.

c. If you recreate the experiment 100 times, 95 times out of 100 the true ATE would fall in the 95% confidence interval.

d. Complete random assignment means treatment will be given to a fixed number of the Group N. Block Random Assignment means that people get grouped into different groups and allocated into Treatment or Control within that group.

e. A balanced design has a smaller standard error.

5.a.

TABLE 2.1

Illustration of potential outcomes for local budgets when village council heads are women or men. [Entries are shares of local budgets allocated to water sanitation.]

Village <i>i</i>	$Y(0)$ Budget share if village head is male	$Y(1)$ Budget share if village head is female	T_i Treatment effect
Village 1	10	15	5
Village 2	15	15	0
Village 3	20	30	10
Village 4	20	15	-5
Village 5	10	20	10
Village 6	15	15	0
Village 7	15	30	15
Average	15	20	5

Possible Combinations, if one Village is assigned to treatment:

V1: $15 - 15.8 = -0.8$

V2: $15 - 15 = 0$

V3: $30 - 14.6 = 15.4$

V4: $15 - 14.6 = 0.83$

$$V5: 20 - 15.8 = 4.2$$

$$V6: 15 - 15 = 0$$

$$V7: 30 - 15 = 15$$

Difference in means: 5

b.

$$5 + 0 + 10 - 5 + 10 + 0 + 15 = 35$$

$$35/7 = 5$$

The true ATE is 5.